A LEFT ADJOINT OF A CONTINUOUS FUNCTOR IS COCONTINUOUS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; every step the author writes out is inside the counted span and no closing sentence is a placeholder.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The description of the pushforward and pullback through a right adjoint. Same chapter, Lemma lemma-have-functor-other-way. Supplies one of the two statements referred to.
- A continuous right adjoint defines the same morphism of topoi. Same chapter, Lemma lemma-have-functor-other-way-morphism. Supplies the other statement referred to.

Source reference. Original lemma 00XX: sources/stacks-sites/pinned-original-source.tex, source lines 5098–5098. Navigation pointer invoked at proof line 5098; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 00XY: sources/stacks-sites/pinned-original-source.tex, source lines 5127–5127. Navigation pointer invoked at proof line 5127; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{C}$ and $\mathcal{D}$ be sites and let $v : \mathcal{D} \to \mathcal{C}$ be a continuous functor. Assume v has a left adjoint $u : \mathcal{C} \to \mathcal{D}$. Then*

(1) *u is cocontinuous,*

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(2) *the results of Lemmas 00XX and 00XY hold.*

In particular, v defines a morphism of sites $f : \mathcal{C} \to \mathcal{D}$.

Proof. Let U be an object of $\mathcal{C}$ and let $\{V_j \to u(U)\}_{j\in J}$ be a covering in $\mathcal{D}$. Then $\{v(V_j) \to v(u(U))\}_{i\in I}$ is a covering in $\mathcal{C}$. Via the adjunction map $U \to v(u(U))$ we can base change this to a covering $\{W_j \to U\}_{j\in J}$ with $W_j = v(V_j) \times_{v(u(U))} U$. Denoting $p_j : W_j \to v(V_j)$ the first projection, we obtain maps

$$u(W_j) \xrightarrow{u(p_j)} u(v(V_j)) \longrightarrow V_j$$

where the second arrow is the adjunction map. This determines a morphism $\{u(W_j) \to u(U)\} \to \{V_j \to u(U)\}$ of families of maps with fixed target, showing that u is indeed cocontinuous. The other statements are immediate from Lemmas 00XX and 00XY. □